

Jaypee Institute of Information Technology

Sector 62



Performance Engineering Lab [17M15CS122]

Project Synopsis

Title: “*Performance Analysis of Load Balancers*”

Submitted To:-

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PROJECT OUTLINE

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INTRODUCTION

As user traffic increases, servers may become overloaded, leading to higher response times and reduced system efficiency. Choosing an appropriate load balancing algorithm is essential for maintaining optimal performance. This project aims to evaluate and compare various load balancing techniques under different workload conditions to determine their effectiveness in improving system performance.

Load balancing is a crucial technique used in modern distributed systems, cloud computing platforms, and web applications to distribute incoming client requests among multiple servers. It helps improve system performance, increase resource utilization, reduce response time, and ensure high availability of services.

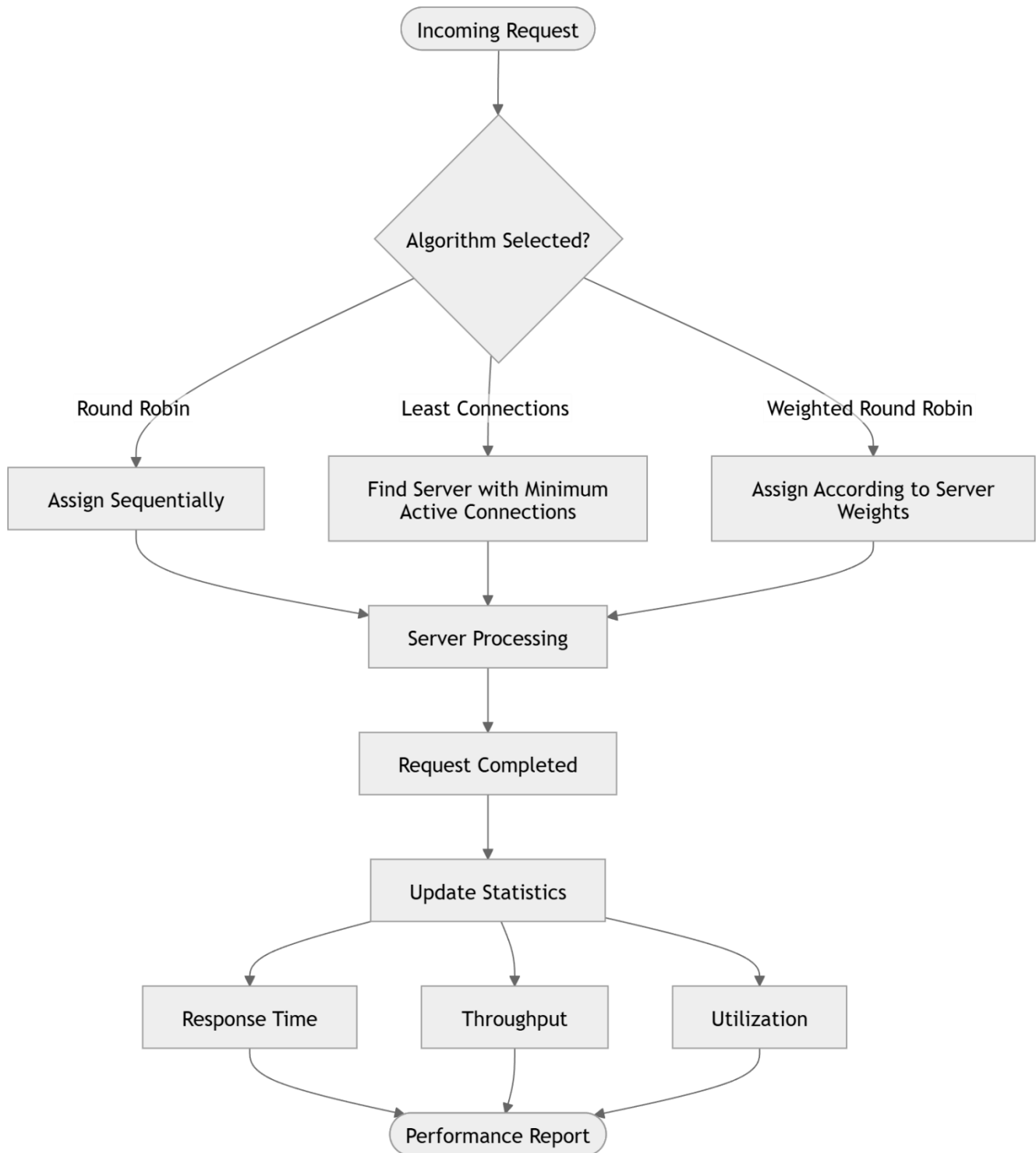
Different load balancing algorithms use different strategies to distribute workloads, which directly affects the overall performance of the system. This project focuses on analyzing and comparing the performance of commonly used load balancing algorithms through simulation.

OBJECTIVES

We have considered below objectives for our project:

- To understand the concept and importance of load balancing in distributed systems.
- To implement and simulate different load balancing algorithms.
- To compare the performance of Round Robin, Least Connections, and Weighted Round Robin algorithms.
- To evaluate system performance using key performance metrics.
- To analyze the effect of increasing workload on server performance.
- To identify the most efficient load balancing strategy for different scenarios

WORKFLOW DIAGRAM



Above diagram explains the workflow of the Load Balancer Performance Analysis system. Incoming requests are first received by the load balancer, which selects a scheduling algorithm such as Round Robin, Least Connections, or Weighted Round Robin. Based on the chosen algorithm, the request is assigned to an appropriate server for processing. After the request is completed, performance statistics including response time, throughput, and server utilization are updated. Finally, these metrics are analyzed and combined to generate a comprehensive performance report for comparing the effectiveness of different load balancing algorithms.

IMPLEMENTATION DETAILS

The system follows a three-tier cloud architecture consisting of a frontend, backend, and database + cloud storage services.

Step 1: Request Generation

Generate client requests at different arrival rates to simulate real-world traffic.

Step 2: Load Balancing

Distribute requests among available servers using:

- Round Robin
- Least Connections
- Weighted Round Robin

Step 3: Request Processing

Servers process assigned requests with varying service times.

Step 4: Data Collection

Record performance statistics during simulation.

Step 5: Performance Evaluation

Analyze and compare results obtained from different algorithms.

Step 6: Visualization

Display results using graphs and charts for better comparison.

****PERFORMANCE METRICS****

Response Time

The time required to process a request from arrival to completion.

Throughput

The number of requests successfully processed per unit time.

Server Utilization

The percentage of time a server remains busy processing requests.

Load Distribution Efficiency

The effectiveness of workload distribution among servers.

TOOLS & LIBRARIES USED

- **Python 3.x** – Used to develop and implement the complete simulation system.
- **SimPy** – Used for discrete-event simulation of servers, requests, and load balancing operations.
- **NumPy** – Used to perform numerical calculations and analyze performance metrics.
- **Matplotlib** – Used to create graphs and visual representations of simulation results.
- **Tkinter (GUI)** – Used to design an interactive graphical user interface for the project.
- **VS Code** – Used as the development environment for programming

EXPECTED OUTCOMES

The outcomes expected to be achieved after developing this project are:

- Comparative analysis of different load balancing algorithms.
- Identification of the algorithm with the best response time.
- Evaluation of throughput under varying workloads.
- Analysis of server utilization patterns.
- Visualization of performance results through graphs.
- Better understanding of distributed system performance optimization.

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